Wazuh SIEM Home Lab - Project Overview

This project documents the design, deployment, and operation of a Wazuh-based Security Information and Event Management (SIEM) home lab. The environment was built to develop practical cybersecurity skills in security monitoring, threat detection, log management, and incident analysis across multiple operating systems.

The lab consists of a Wazuh Manager and agents deployed on Kali Linux, Ubuntu, and Windows virtual machines. In this environment, I performed various security-related activities to generate and analyze alerts, including user account creation, failed login attempts, privilege escalation events, Sudo activity, account modifications, and other security-relevant actions. These events were collected and correlated within Wazuh to provide visibility into endpoint activity and demonstrate foundational Security Operations Center (SOC) workflows.

The project also incorporates MITRE ATT&CK concepts to help understand attacker behaviors and how security monitoring tools can detect activity associated with commonly observed tactics and techniques. This lab strengthens hands-on experience with SIEM technologies, threat detection, log analysis, incident investigation, and cybersecurity operations while building a portfolio of practical security projects.

Key Skills Demonstrated

- SIEM Deployment and Administration
- Security Monitoring and Alert Analysis
- Threat Detection and Investigation
- Log Collection and Correlation
- MITRE ATT&CK Framework Mapping
- Linux and Windows Security Monitoring
- User and Account Activity Auditing
- Security Operations (SOC) Concepts
- Incident Detection and Response

Technologies Used

- Wazuh SIEM
- Kali Linux
- Ubuntu Linux
- Windows
- VirtualBox
- MITRE ATT&CK Framework